

HANUMAN KUMAR SINGH

Java Spring Boot Developer | AI Engineer (Spring AI, RAG) | Backend & Cloud Engineer

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PROFESSIONAL SUMMARY

Java Spring Boot Developer with 3+ years building scalable microservices, event-driven architectures, and cloud-native REST APIs for enterprise SaaS and IoT platforms; now expanding into applied AI Engineering with Spring AI, Retrieval-Augmented Generation (RAG), and LLM integration. Expert in Spring Boot, Spring Security, Spring Data JPA, Hibernate, Kafka, and RabbitMQ, with systems handling 50,000+ daily transactions and 1,000+ IoT device integrations. Hands-on with OpenAI, AWS Bedrock, and MCP for RAG pipelines, vector embeddings, AI agent tool-calling, and AI observability. Proficient in Docker, Kubernetes, AWS, and CI/CD. Full SDLC ownership in high-velocity Agile environments.

TECHNICAL SKILLS

Languages: Java 8+ (Streams, Lambdas, Multithreading, Concurrency), JavaScript, TypeScript, SQL, Python, Dart

Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Spring Cloud, Node.js, Express.js, REST APIs, Microservices

AI / Generative AI: Spring AI, RAG, OpenAI API, AWS Bedrock, Model Context Protocol (MCP) Client & Server, Prompt Engineering, Embeddings & Vector Stores, AI Agents & Tool Calling, Chat Memory/Advisors, AI Evaluators & Testing, AI Observability

Messaging & Data: Apache Kafka, RabbitMQ, AWS SQS/SNS, MySQL, PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch, CloudFormation), Docker, Kubernetes, Helm, Jenkins, GitHub Actions, CI/CD, Linux

Testing & Security: JUnit 5, Mockito, TestContainers, REST Assured, OAuth2, JWT, Spring Security

Patterns & Frontend: CQRS, Event Sourcing, Saga, API Gateway, SOLID; Angular, React.js, Redux, HTML5/CSS3, Flutter (iOS & Android)

Tools: Git, Maven, Gradle, Swagger/OpenAPI, Prometheus, Grafana, Postman, Agile/Scrum

PROFESSIONAL EXPERIENCE

Smartly Built | Software Developer Feb 2024 - April 2026

- Engineered Spring Boot REST APIs (Spring Data JPA, Hibernate, Spring Security RBAC) for JackTrade CRM (500+ screens, 15+ tenants); cut API response time 35% via query optimization and multi-tenant data isolation.
- Integrated Kafka for real-time IoT event streaming and RabbitMQ for async processing on Greyter Water Systems; added Redis caching, cutting latency ~40% across 50,000+ daily device requests.
- Automated CI/CD with Jenkins, Docker, and Kubernetes on AWS; deployed Prometheus/Grafana/CloudWatch observability, achieving 99.9% uptime. Prototyped Spring AI (OpenAI, RAG, MCP) integrations for AI-assisted features.

Indoc Web Solutions Pvt Ltd | Software Developer Mar 2023 - Jan 2024

- Developed Spring Boot RESTful APIs (Spring Data JPA, MySQL/PostgreSQL) for mobile backends; tuned Hibernate queries and DB indexes, cutting response time ~30%.
- Shipped 3 production Flutter apps (iOS & Android) with Provider/BLoC state, Firebase, and third-party SDKs, achieving <0.5% crash rate.

Mindful Machines Pvt Ltd | Software Developer Apr 2022 - Feb 2023

- Built Spring Boot RESTful APIs and Node.js services for Urban Kirana & Near Bajar; optimized JPA/Hibernate queries and MongoDB aggregations for 50,000+ daily active users.
- Contributed to distributed system design and Agile workflows; delivered 100% of assigned features on schedule across concurrent projects.

KEY PROJECTS

JackTrade CRM - Enterprise Multi-Tenant Platform | Spring Boot, Spring Security, Angular, MongoDB, Redis, RabbitMQ, Docker, AWS

- 50+ Spring Boot REST APIs with RBAC for multi-tenant isolation across 15+ clients; RabbitMQ + Redis cut API latency 40% across 50,000+ daily requests. 500+ screen Angular front end; 99.9% uptime via Docker/Jenkins CI/CD on AWS.

Greyter Water Systems - IoT Device Management Platform | Spring Boot, Angular, AWS IoT Core, MQTT, Kafka, Redis, Prometheus

- Spring Boot APIs processing 50,000+ daily MQTT messages from 1,000+ IoT devices via AWS IoT Core; Kafka telemetry streaming, Redis caching, RBAC device access, and an automated OTA firmware pipeline for 500+ devices.

Spring AI RAG & Agent Prototypes | Spring AI, OpenAI, RAG, MCP, Vector Embeddings, Prompt Engineering

- Built RAG pipelines with Spring AI using vector embeddings and prompt stuffing to ground LLM responses; implemented AI agent tool-calling and MCP client/server integrations, with AI observability and evaluators for response quality.

EDUCATION

B.Tech, Computer Science & Engineering - University of Engineering and Management, Jaipur | CGPA: 7.80/10 **2019 - 2023**

CERTIFICATIONS

- From Java Dev to AI Engineer: Spring AI Fast Track - Udemy (2026) - Spring AI, OpenAI, RAG, MCP, AI Agents, AI Testing & Observability, Speech & Image Generation
- AWS Certified Developer - Associate (In Progress) | Java Spring Professional - Udemy | Cloud Computing & DevOps - Intellipaat (2023)